HDMI EXTENSION BOARD — XEM8310 CONNECTIONS, POWER AND MOUNTING
MC1
BTE–040–01–F–D–A
MC2
Power Jack
12 V nominal, center positive
J4
PJ–063AH
TP710
12V_IN
+VDCIN
XEM8310 MODULE MOUNTING HOLES
MH1
MH2
FRONTEND 5 V POWER | CUSTOM MICRO–HDMI PORTS – NOT STANDARD HDMI
U700
TPSM63606RDLR
PORT_ENABLE
PCIe x4 — BRK8310 reference circuit
U2
PCA9306DCUR
USB-C JTAG PROGRAMMER, SELECTION AND FRONTEND PORTS
USB TO JTAG PROGRAMMER
J500
USB4110–GF–A
U501
FT232HQ–REEL
U505
93LC56BT–I/SN
Y500
1.8 V JTAG TRANSLATOR
U602
TXU0304PWR
CUSTOM FRONTEND INTERFACE – NOT HDMI VIDEO
FRONTEND PORT 1 | JTAG SWITCH + MICRO–HDMI
FRONTEND PORT 2 | JTAG SWITCH + MICRO–HDMI
FRONTEND PORT 3 | JTAG SWITCH + MICRO–HDMI
FRONTEND PORT 4 | JTAG SWITCH + MICRO–HDMI
FRONTEND PORT 5 | JTAG SWITCH + MICRO–HDMI
FRONTEND PORT 6 | JTAG SWITCH + MICRO–HDMI
FRONTEND PORT 7 | JTAG SWITCH + MICRO–HDMI
FRONTEND PORT 8 | JTAG SWITCH + MICRO–HDMI
U611
TS3A27518EPWR

1. Function: ties VREFP (MC2.79) to GNDADC (MC2.77), which selects the FPGA internal ADC reference.

2. Source / changes: Opal Kelly XEM8310 Hardware Design Guide, Selecting the ADC Reference Voltage, revision not stated:
https://docs.opalkelly.com/xem8310/hardware–design–guide/
BRK8310–A01 SYSMON Header R6 = 0 ohm is a direct wire here.
ADC readings are not verified on hardware.

Function: XEM8310 connectors provide the board power, FPGA I/O, JTAG and GT connections; J4 is the 12 V input and MH1/MH2 mount the module.

Source / changes: Opal Kelly BRK8310–A01 p.1 (2022–05–19): https://pins.opalkelly.com/downloads/442/download?category_id=3 ; Codex assigned the custom HDMI and x4 nets. VIO1 target is 1.8 V: program and verify it. V25 command–pair remap (Howard, 3 Nov 2026): port –> Bank 64 pair 1–>L5, 2–>L6, 3–>L3, 4–>L7, 5–>L4, 6–>L1, 7–>L2, 8–>L8 so all 8 pairs stay on L3 without crossings; update the FPGA XDC to match. V24: J4 changed from CUI PJ–102AH (5 A) to Same Sky PJ–063AH (8 A, 24 V, same 2.0 mm pin and 5.5 mm plug, no switch pin; unused PJ–102AH pin 3 dropped): https://www.sameskydevices.com/product/resource/pj–063ah.pdf ; footprint = KiCad library BarrelJack_CUI_PJ–063AH_Horizontal, pin 1 at the old pin 1. Mating plug and 12 V adapter must also be rated above 5 A; footprint not re–measured against the drawing. V29 (Howard, 4 Nov 2026): BRK8310 standoff holes MH3/MH4 beside MC3 removed for routing space; the XEM8310 is now held by standoffs at MH1/MH2 only and by its connectors at the MC3 end (mechanical retention not verified).

Function: U700 converts +VDCIN to FE_5V (5.02 V nominal, 6 A module rating).
PG and R703 enable all eight ports once the central rail is in regulation.

Source / changes: TI TPSM63606 SLVSGB4B Rev. B, Fig. 9–1 and Table 8–1:
https://www.ti.com/lit/ds/symlink/tpsm63606.pdf . Codex adapted prior V14 U100:
R700 40.2k, R701 10k, R702 13k (all 1%), C708 22pF set 5 V / 1 MHz; PG gates switches.
C700–C703 10uF / 50V; C704–C707 use 47uF / 16V X5R instead of prior X6S.
CBOOT/RBOOT shorted, VLDOIN tied to output. Check effective capacitance,
loop stability, input damping, startup, thermal performance and PCB layout.

Function: U701–U708 individually switch FE_5V to J610–J617 pin 19; red LEDs indicate each switch FAULT. TP701–TP708 access the outputs. All ports start together: no staggering.
Source / changes: TI TPS2553 SLVS841G Rev. G, pins / sections 8–9 and Table 9–2: https://www.ti.com/lit/ds/symlink/tps2553.pdf . Codex adapted prior V14: EN now follows U700 PG:
R711–R718 = 37.4k 1% (630–771 mA limit); C711–C718 are IN bypass; V31: C721–C728 1 uF (Samsung CL10A105KB8NNNC, LCSC C15849) added on each OUT / PORTn_5V per SLVS841G sec. 9.2.1.2.4 p. 21 and the p. 1 ESD rating with external capacitance; value chosen by Claude, not given by TI: R721–R728 / D701–D708 added as FAULT lights. Each OUT goes to a separate pin–19 net.
Cable current and return rating, hot–plug / ESD, backfeed, 8–port inrush and thermal behavior are not verified. Existing 3.3 V–only frontends must NOT be connected to these 5 V ports.

Function: Connects PCIe x4 data and 100 MHz reference clock, with reset, SMBus and WAKE#. Host slot 12 V is isolated from the board 12 V input.

Source / changes: Opal Kelly BRK8310–A01 schematic, p. 2, 2022–05–19: https://pins.opalkelly.com/downloads/442/download?category_id=3
Adapted from x8 to x4: fit R28 = 0 ohm; omit R29 and unused lanes 4–7 and C21–C26/C29–C30. R2/R7 and R9–R11 retained.
Unused options R1/R8, R16/R17 and R27 are omitted.
Host +12 V goes only to TP20. Host 3.3 V and AUX power the original sideband circuitry; they do not feed the main 12 V input.
PCA9306 Rev O, Fig. 8–1, section 10: https://www.ti.com/lit/ds/symlink/pca9306.pdf : SN74AXC1T45 Rev E: https://www.ti.com/lit/ds/symlink/sn74axc1t45.pdf
AP22653 DS41186 Rev 5–2, sections 7–9: https://www.diodes.com/datasheet/download/AP22652_AP22653_AP22652A_AP22653A.pdf
Q1 Si1922EDH dual FET, product datasheet: https://www.vishay.com/en/product/67192/
V30 (Howard, 4 Nov 2026): TX AC–coupling caps C13–C20 changed from BRK8310's 0.1 µF (Kemet C0402C104K8PACTU) to 0.22 µF (Samsung CL05B224KO5NNNC, LCSC C16772, 0402 X7R 16 V); same footprint and routing.
Reason: PCIe Gen3 TX coupling range 176–265 nF (commonly cited PCIe Base Spec value, also in AMD UG583; the PCI–SIG document itself was not read); 220 nF also covers Gen1/2. C7/C8 (REFCLK) unchanged.
Working target: VIO1 = 1.8 V (Howard, 3 Nov 2026; needed for PCA9306 >= 1.2 V and Bank 64 LVDS); set in XEM before testing and verify all Bank 64 I/O. Factory default 1.0 V is below PCA9306 range: https://docs.opalkelly.com/xem8310/device–settings/
ERC power flags at U2/U6 VREF2 represent their R13/R23 200k resistor–fed bias, not independent regulators. Confirm bias and startup.
PCIe/HDMI combined Vivado design, clock, VIO1 choice, host sequencing and signal integrity remain to be verified.

Function: J500 and U501 convert USB–C to JTAG; U505 stores the programmer settings.
U602 translates the JTAG lines to 1.8 V. SW600 selects the XEM or one of eight frontends.
U603 checks the selected target voltage; U605 enables the translator after a delay.
C601/C603 bypass the XEM rails; U610 selects the XEM path. Turn OFF before selecting.

Source / changes: USB–C input: Codex adaptation of BRK8305 A03,
25Oct2023, JTAG sheet8/PDFp4: https://pins.opalkelly.com/downloads/583/download?category_id=3
J500=USB4110–GF–A; D500=USBLC6–2SC6. C506
changed 10nF to 100nF. USB shell uses C502/FB503;
shared DGND. Connector fit and USB/ESD layout unverified.
ST USBLC6–2 Rev7 Fig15: https://www.st.com/resource/en/datasheet/usblc6–2.pdf

FT232H and EEPROM: Codex adaptation of BRK8305 A03,
25Oct2023, JTAG sheet8/PDFp4: https://pins.opalkelly.com/downloads/583/download?category_id=3
FT232HQ pins13/14/15/16=TCK/TDI/TDO/TMS; ADBUS4–7
and ACBUS NC as BRK8305. Former R508 removed.
U505=93LC56BT–I/SN (SOIC) replaces CFG001/MSOP;
Y500=SJK replaces Abracon; FB500–502=Murata BLM18AG601SN1D.
Y500 specifies 18pF load; 18pF C503/C504 give ~9pF + stray.
Confirm crystal load, frequency, drive and startup after layout.
Y500: https://xonstorage.z8.web.core.windows.net/pdf/sjk_7u12000e18ucg_xonjuly20_20_link.pdf
Bead current and voltage drop need checks.
Microchip DS21794F Table3–1: https://ww1.microchip.com/downloads/en/DeviceDoc/21794F.pdf
FT232H DS v2.2 sec3,6: https://ftdichip.com/wp–content/uploads/2024/09/DS_FT232H.pdf
AMD program_ftdi EEPROM write/readback and Vivado
enumeration must be tested. VCK190 match not established.
https://docs.amd.com/r/2024.1–English/ug908–vivado–programming–debugging/Programming–FTDI–Devices–for–Vivado–Hardware–Manager–Support

Translator: Codex adaptation; TI TXU0304 Rev A sec.9: https://www.ti.com/lit/ds/symlink/txu0304.pdf
VCCA = FTDI3V3; VCCB = XEM +1.8 V (MC1.2); R601-R603 = 22R provisional.
R604 pull–up removed; IC inputs have weak pulldowns. Check rail sequencing, unpowered–target
injection, timing, cable loading and FTDI/Vivado operation.

Rail bypass: Codex adaptation: C601/C603 retained from the removed regulators.
XEM8310 MC2.70 = +3.3 V and MC1.2 = +1.8 V; rail budget and startup must be verified.
Opal Kelly XEM8310 pin list: https://pins.opalkelly.com/pin_list/XEM8310

Rotary selector: Codex direct–to–enable wiring; nine 10k pulls replace decoder logic.
ALPS SRBQ 1P9T non–shorting; confirm stock, contact sequence and footprint.
https://tech.alpsalpine.com/e/products/detail/SRBQ490100/
Bounce and switch faults are not proven.

OFF/ENABLE: Codex dual–pole arrangement; R600 biases MASTER low.
C&K JS datasheet p.9; verify contact positions and footprint:
https://www.littelfuse.com/assetdocs/littelfuse–c–k–slide–js–series–datasheet?assetguid=aba42b08–0d2c–423b–813d–a2faa5a3bb14

Voltage window: Codex circuit; TI TPS3700 Rev G sec.7.3: https://www.ti.com/lit/ds/symlink/tps3700.pdf
XEM 3.3 V supplies U603. Selected VREF uses XEM R629 or HDMI pin 1.
R605-R608=301k/100k and 402k/100k; R613/C608 discharge/filter.
Nominal window 1.604-2.008 V; verify leakage, tolerance and startup.

Delay: Codex circuit; TI TPS3808 Rev N secs.4, 7: https://www.ti.com/lit/ds/symlink/tps3808.pdf
U605 changed G33 to G30: 2.79 V nominal threshold increases 3.3 V rail margin.
CT open: nominal 20 ms; R615=12k now pulls to XEM 3.3 V, R616=100k pulls down.
Startup and brownout remain unverified; keep OFF during power–up.
JLC C189211 listing conflicts with TI G30 threshold; verify MPN/marking.

XEM target switch: Codex circuit using TI TS3A27518E Rev F Fig.5–2, Table 8–1:
https://www.ti.com/lit/ds/symlink/ts3a27518e.pdf
IN1/IN2 high select NO; active–low EN from rotary; 10k pullup. Channels (Howard 1 Nov 2026, for straight PCB routing): TCK=4, TDI=6, TMS=3, TDO=5, VREF=2; channel 1 unused.
NO2–NO6: ±6 kV IEC contact rating; not a 5 V DC clamp.
Selection, leakage, power–off and SI need checks.
XEM JTAG pins: https://docs.opalkelly.com/xem8310/expansion–connectors/

Function: HDMI1 selects shared JTAG and VREF.
100R series parts alone do not provide 5 V protection.

Source / changes: Codex circuit using TI TS3A27518E Rev F Fig.5–2, Table 8–1:
https://www.ti.com/lit/ds/symlink/ts3a27518e.pdf
IN1/IN2 high select NO; active–low EN from rotary; 10k pullup. Channels (Howard 1 Nov 2026, for straight PCB routing): TCK=4, TDI=6, TMS=3, TDO=5, VREF=2; channel 1 unused.
NO2–NO6: ±6 kV IEC contact rating; not a 5 V DC clamp.
Selection, leakage, power–off and SI need checks.

V24 one–page schematic: J4 = PJ–063AH

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Title: HDMI Extension Board – XEM8310 and USB–C JTAG

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